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SHRESTH BIPIN

Software Engineer · Full-Stack, Cloud & Applied AI

SUMMARY

Computer Science undergraduate (batch of 2027) who builds products end to end: React/Next.js and React Native front ends, FastAPI, Django and Node.js back ends, SQL and NoSQL data layers, and cloud deployment with CI/CD. Has shipped GenAI/RAG systems in production, written CUDA kernels with 100×+ speedups, and presented technical trade-offs to senior leadership.

EXPERIENCE

AI/ML Intern — Motherson Technology Services, Noida

Jun 2026 – Jul 2026

Python · FastAPI · React 19 · TypeScript · PostgreSQL · Docker · Azure DevOps · CI/CD

- Built **CapEx Intelligence**, a GPT-4o + RAG service over a 5-plant, ~5,000-line-item capital-expenditure dataset; compared RAG, keyword and hybrid retrieval, then shipped it as a FastAPI + React 19/TypeScript app on Azure with CI/CD.
- Built an **85%-confidence QA gate** inside a 29-component production pipeline, routing low-confidence outputs to human review before they reached SAP/MES.
- Benchmarked three real-time inference options (Gemini Live API, GPT-4o Realtime, LiveCC-7B) on accuracy, latency, cost and on-prem feasibility; the report set the production direction.
- Scored four architecture stacks on security, scalability, fidelity and cost (recommendation adopted); documented 60+ pipeline components and presented trade-offs to senior plant leadership.

PROJECTS

VORI — Cloud-Native E-Commerce Platform

UI demo: vori-microservices-ecommerce.vercel.app

Django · FastAPI · Node.js · React · PostgreSQL · Redis · Pinecone · Docker

- Architected three independently containerized microservices (auth/payments, semantic search, cart cache) with CI/CD on Render and Vercel; a Redis cache cut cart-retrieval latency ~80% versus direct database queries in benchmarks.

AWS Route 53 Clone

github.com/shresth135/route53-clone

Next.js · Tailwind CSS · FastAPI · SQLAlchemy · Pydantic · Vercel · Render

- Full-stack DNS management app: hosted zones and 9 record types (A, AAAA, CNAME, TXT, MX, NS, PTR, SRV, CAA) with real-time search, served by a validated FastAPI REST API with live Swagger docs.

E-Commerce AI Shopping Assistant

github.com/shresth135/ecommerce-chatbot

Python · Streamlit · FAISS · Sentence Transformers · Google Gemini

- RAG chatbot that searches, recommends and compares products and answers store FAQs, using intent detection, FAISS vector search over sentence embeddings, and conversation memory.

Parallel Image Processing — GPU Acceleration

github.com/shresth135/cuda-image-processing

CUDA · C · C++

- Wrote CUDA kernels for Sobel edge detection and Gaussian blur using shared memory; cut 1024×1024 processing from 112.7 ms to 0.8 ms (135.6× speedup), 100.7× at 512×512 and 65× at 4096×4096.

Mobile App Migration

React Native (Expo) · FastAPI · MongoDB

- Migrated a full-stack mobile app off the Emergent AI platform to a local development environment, with app-store release through EAS build/submit as the next step.

TECHNICAL SKILLS

Languages	Python, TypeScript/JavaScript, Java, C/C++, SQL, R
Web & Mobile	React, Next.js, React Native (Expo), Node.js, FastAPI, Django REST Framework, Streamlit, Tailwind CSS, REST APIs
Data & AI	PostgreSQL, MongoDB, Redis, SQLAlchemy, Pinecone, FAISS, RAG, GPT-4o, Gemini, Sentence Transformers, Power BI
Cloud & DevOps	Azure, Azure DevOps, AWS, Docker, CI/CD, Vercel, Render, Git/GitHub
Systems	CUDA, parallel computing, data structures & algorithms, operating systems, DBMS

EDUCATION

Manipal Institute of Technology, Manipal — BTech, Computer Science & Engineering

2023 – 2027

Coursework: DSA, DBMS, Operating Systems, Computer Architecture, Machine Learning, Compiler Design, Parallel Computing

CERTIFICATIONS

AWS Cloud Practitioner Essentials (Aug 2026) · Microsoft Excel VBA and Macros (Aug 2026) · Microsoft Python Development (May 2026) · Microsoft Power BI Data Analyst (Apr 2026) · IBM Data Analytics with Excel & R (Apr 2026)

ACHIEVEMENTS & ACTIVITIES

Top 240 district-wide, IOQM · Merit Certificate, ANCQ · Qualified Stage 1, NTSE · Fine Arts (4th-year diploma) · Vocal Music (6th-year diploma) · Brown IV Kyu, Shito-Ryu Karate-Do